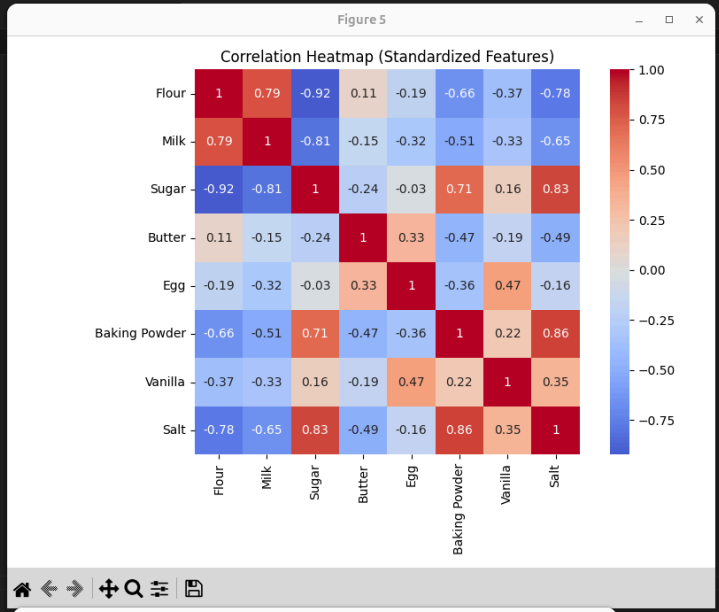
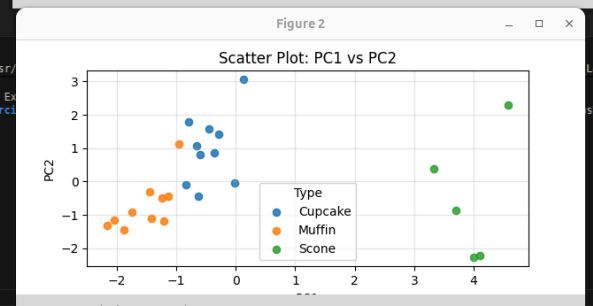
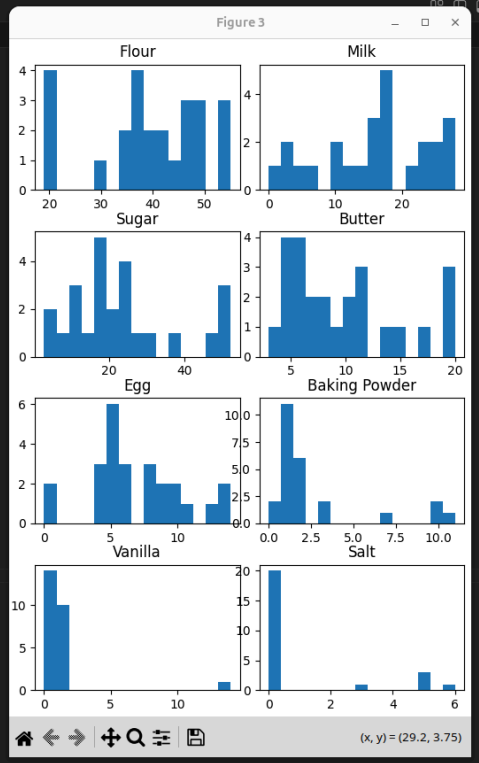
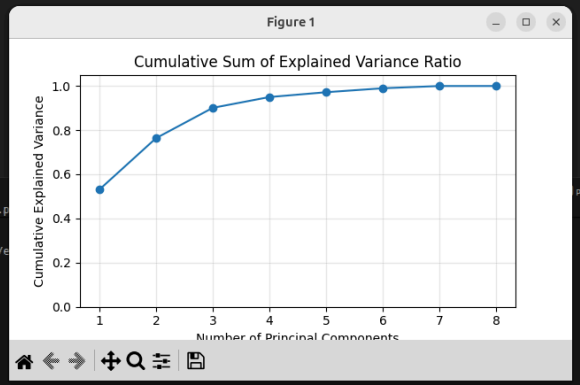
Feb 6 21:40

File Edit Selection View Go Run Terminal Help

exercise1.py | recipes_muffins_cupcakes_scones.csv

Exercise1 > exercise1.py > ...

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

df = pd.read_csv("recipes_muffins_cupcakes_scones.csv")

num_df = df.select_dtypes(include=[np.number]).copy()

label_col = None
for c in df.columns:
    if c not in num_df.columns and df[c].nunique() <= max(2, min(10, len(df) // 5)):
        label_col = c
        break

X = num_df.values
scaler = StandardScaler()
Xz = scaler.fit_transform(X)

pca = PCA(n_components=8, random_state=0)
Z = pca.fit_transform(Xz)

evr = pca.explained_variance_ratio_
cum = np.cumsum(evr)

print("Explained variance ratio (PC1..PC8):")
print(evr)
print("\nCumulative explained variance (PC1..PC8):")
print(cum)

plt.figure()
plt.plot(range(1, len(cum) + 1), cum, marker="o")
plt.xticks(range(1, len(cum) + 1))
plt.ylim(0, 1.05)
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("Cumulative Sum of Explained Variance Ratio")
plt.grid(True, alpha=0.3)

plt.figure()
if label_col is not None:
    labels = df[label_col].astype(str).values
    for lab in np.unique(labels):
        m = labels == lab
        plt.scatter(Z[m, 0], Z[m, 1], s=35, alpha=0.85, label=lab)
    plt.legend(title=label_col, frameon=True)
else:
    plt.scatter(Z[:, 0], Z[:, 1], s=35, alpha=0.85)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("Scatter Plot: PC1 vs PC2")
plt.grid(True, alpha=0.3)

plt.figure(figsize=(10, 10))
cols = num_df.columns.tolist()
n = len(cols)
r = int(np.ceil(n / 2))
for i, c in enumerate(cols, 1):
    ax = plt.subplot(r, 2, i)
    ax.hist(df[c].dropna().values, bins=15)
    ax.set_title(c)
plt.tight_layout()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_3/WilliamsBenjamin_Lab3$ /usr/
FileNotFoundError: [Errno 2] No such file or directory: 'recipes_muffins_cupcakes_scones.csv'
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_3/WilliamsBenjamin_Lab3$ cd Ex
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_3/WilliamsBenjamin_Lab3/Exerci
Explained variance ratio (PC1..PC8):
[5.30713818e-01 2.32926199e-01 1.37271639e-01 4.89399359e-02
 2.18813363e-02 1.79333825e-02 1.02410755e-02 9.26139804e-05]

Cumulative explained variance (PC1..PC8):
[0.53071382 0.76364002 0.90091166 0.94985159 0.97173293 0.98966631
 0.99990739 1.        ]

PC1 highest positive loading feature: Salt 0.4556096741864202
PC1 most negative loading feature: Flour -0.4452797934996037
PC2 highest positive loading feature: Egg 0.6628137748507001
PC2 most negative loading feature: Milk -0.3233860340000608
```